



PLANT: ECONOMIC IMPORTANCE — FULL MIND MAP

Biology Ch. 8.v · Plant Physiology · Hinglish · SSC / BPSC / BSSC / Railway



Topics in this sub-chapter:

1. Important Crop Varieties & Botanical Names
2. Medicinal, Industrial & Useful Plants
3. Palm Oil, Canola, Moringa & Native Plants
4. Biofuels, Petro-crops, Jatropha & Ethanol
5. Chilgoza, Neem & Medicinal Plant Uses
6. Caffeine — Cola, Tea & Coffee
7. Seaweed, Spices, Oils & Plant Products
8. Quinine, Cinchona & Anti-malarial Drugs
9. Cork & Cotton
10. Opium Poppy, Morphine, Heroin & Alkaloids
11. Tapa Cloth — Moraceae
12. Laminaria, Iodine & Goitre

1. Important Crop Varieties & Botanical Names

- Kanchan, Krishna aur Banarasi — Amla/Indian gooseberry ki improved varieties hain; Amla vitamin C aur medicinal properties ke liye noted hai. [Q1]
- Neelum × Alphonso mango cross se Ratna hybrid variety bani; source ise Maharashtra aur Karnataka ke kuch parts mein grown high-yielding mango batata hai. [Q2]
- Kashi Lalima red okra/lady finger variety hai, IIVR Varanasi ke experts dwara developed; source antioxidants, calcium aur iron richness highlight karta hai. [Q3]
- Co. 1148 eastern Uttar Pradesh ke liye important sugarcane variety hai; medium-thick canes, good germination/tillering aur moderate drought resistance source features hain. [Q4]
- Crop–variety match mein Mustard–Varuna, Field Pea–Sapna aur Groundnut–Kaushal correct hain; Linseed–Surya source ke mutabik incorrect pair hai. [Q5]
- Macaroni wheat ka botanical name Triticum durum hai; source ise tetraploid wheat species batata hai. [Q6]

2. Medicinal, Industrial & Useful Plants

- Parthenium hysterophorus leaf extract source mein biologically synthesized nanoparticles/cancer-treatment potential se linked hai aur anti-cancer medicinal attributes mention kiye gaye hain. [Q7]
- Litmus acid-base indicator lichens se extracted water-soluble dyes ka mixture hai; filter paper par absorb karke acidity/alkalinity test ki jati hai. [Q8]
- Neem oil: pesticide/insect-mite control aur pharmaceutical uses — dono correct; hospital detergent manufacture ke liye neem seeds wale statement ko source unsupported/incorrect maanta hai. [Q9]
- Pine, Fir, Spruce, Cedar, Larch aur Cypress timber-yielding gymnosperms hain; source 'naked seeds' feature ko contrast karta hai angiosperms ke enclosed ovules/seeds se. [Q10]
- Mushrooms ke medicinal, psychoactive, insecticidal aur bioluminescent — chaaron properties source mein correct hain. [Q11]
- Aloe vera medicinal crop hai; source antiviral/antibacterial properties aur medicinal/cosmetic use mention karta hai. [Q12]
- Tulsi (Ocimum tenuiflorum / O. sanctum) ki medicinal value source ke mutabik phenols aur flavonoids सहित secondary metabolites se linked hai. [Q13]

3. Palm Oil, Canola, Moringa & Native Plants

- Palm oil ke source statements mein lipstick/perfume industry use aur biodiesel use correct hain; principal African oil palm *Elaeis guineensis* West/Southwest Africa native hai, isliye Southeast Asia-native statement incorrect hai. [Q14]
- Source ke National Policy on Biofuels-2018 framing mein biofuel raw materials: cassava, damaged wheat grains, rotten potatoes aur sugar beet; groundnut seeds aur horse gram listed nahi. [Q15]
- Canola = human consumption ke liye bred oilseed mustard/rapeseed type; key feature source mein very low erucic acid content hai. [Q16]
- Moringa source ke mutabik leguminous nahi; tamarind South Asia endemic nahi. Correct statements: India mein tamarind largely minor forest produce ke roop mein collect hota hai, India tamarind/moringa seeds export karta hai, aur dono ke seeds biofuel production mein use ho sakte hain. [Q17]
- Cashew, Papaya aur Red Sanders mein source ke mutabik sirf Red Sanders India-native/endemic hai; cashew Brazil aur papaya tropical America origin se linked hain. [Q18]

4. Biofuels, Petro-crops, Jatropha & Ethanol

- Fourth-generation biofuels genetically modified microorganisms—source example algae, yeast, fungus aur cyanobacteria—se banaye ja sakte hain; question answer GM algae hai. [Q19]
- Pongam/Pongamia pinnata ek leguminous petroplant hai; seeds se oil milta hai jo biodiesel use se linked hai. [Q20]
- Petro-crops hydrocarbons aur lipids se rich hote hain; source Jatropha ko petro-crop example deta hai. [Q21]
- Source ke historical question ke mutabik India ka first biodiesel plant Kakinada, Andhra Pradesh mein establish hua. [Q22]
- Jatropha curcas ko bio-diesel plant / future biodiesel feedstock kaha gaya hai; harsh environment mein grow, non-edible crop aur seeds biodiesel ke liye use hote hain. [Q23, Q24, Q25]
- Sugarcane juice/molasses fermentation se ethanol biofuel milta hai; molasses ko source power alcohol ka excellent raw material batata hai. [Q26, Q27]

5. Chilgoza, Neem & Medicinal Plant Uses

- Chilgoza *Pinus gerardiana* (pine) ke seeds se milta hai; source carbohydrate aur protein richness mention karta hai. [Q28]
- Chief mosquito repellent source answer Neem/Mango hai; neem ke seeds/leaves medicinal use se linked hain. [Q29]
- Medicinal-use matching: Brahmi → Brain tonic | Sadabahar → Anti-diabetic | Pudina → Carminative | Tulsi → Expectorant. [Q30]

6. Caffeine — Cola, Tea & Coffee

- Cola-type soft drinks mein significant caffeine hota hai; caffeine xanthine alkaloid stimulant hai. [Q31]
- Caffeine naturally tea aur coffee dono mein milta hai; source cocoa/cola/soft drinks bhi mention karta hai. [Q32]
- Tea leaves ka most important stimulant caffeine hai; source ise widely consumed CNS stimulant/psychoactive drug batata hai. [Q33]

7. Seaweed, Spices, Oils & Plant Products

- Seaweeds iodine ka important/rich natural source hain. [Q34]

- Camphor, Chicory aur Vanilla — source ke mutabik teeno plant products hain: camphor laurel wood, chicory root use, aur Vanilla orchids se flavour. [Q35]
- Kuttu flour Buckwheat/Fagopyrum esculentum se milta hai; source ise pseudocereal batata hai. [Q36]
- Saffron Crocus sativus flower ke stigma/styles ('threads') se banta hai. [Q37]
- Peppermint/Mentha arvensis mein oil ka maximum source answer leaf hai; peppermint/menthol oil leaves ki steam distillation se banta hai. [Q38]
- Source Q39 ka answer Asafoetida/Hing hai; explanation ke mutabik ye Ferula species ka dried oleo-gum resin hai jo rhizome aur fleshy tap roots se extracted hota hai. [Q39]
- Product–source match: Opium → Fruit | Hing → Root | Rubber → Stem | Quinine → Bark. [Q40]

⚠ **EXAM TRAP — Hing / Asafoetida Source Question**

- ▶ Q41 mein source 'Multiple Correct Answers (Ans. A & B)' mark karta hai: stem exudation aur roots extraction dono accepted; explanation rhizome (modified stem) + fleshy tap roots se dried sap/oleo-gum resin batata hai. [Q41]

8. Quinine, Cinchona & Anti-malarial Drugs

- Quinine malaria drug Cinchona tree se milta hai; Cinchona source ke mutabik angiosperm hai. [Q42]
- Quinine ka source part Cinchona bark/stem bark hai — ye same fact multiple exam questions mein repeat hua hai. [Q43, Q44, Q45, Q46]
- Arteether medicine ko source seeded plant Artemisia annua se obtained batata hai; malaria Plasmodium parasite se linked hai. [Q47]
- Natural quinine ke replacement ke roop mein source synthetic drug Chloroquine batata hai. [Q48]

⚠ **EXAM TRAP — Hydroxychloroquine**

- ▶ Q49 source mein disputed/wrong question hai: hydroxychloroquine kisi plant bark se obtained nahi; explanation ise synthetic drug batata hai, Cinchona-derived natural quinine se alag. [Q49]

9. Cork & Cotton

- Cork oak Quercus suber ki bark se cork milta hai; source Mediterranean origin aur waxy cork cells mention karta hai. [Q50]
- Cotton ka major component cellulose hai; source chemical composition mein cellulose ~91% diya gaya hai. [Q51]

10. Opium Poppy, Morphine, Heroin & Alkaloids

- Noscapine poppy (Papaver somniferum) se milta hai aur source ise benzyloquinoline alkaloid batata hai. [Q52]
- Heroin opium poppy se obtained hoti hai; opium latex ke morphine ko chemically process karke heroin ban sakti hai. [Q53]
- Morphine source ke mutabik poppy ke unripe fruit se acquired hoti hai aur CNS par directly act karti hai. [Q54]
- Opium = unripe poppy fruit/capsule se milne wala dried milky latex; shallow cuts se latex collect kiya jata hai. [Q55, Q56]
- Opium ka main constituent source mein ~12% morphine diya hai. [Q57]
- Opium poppy ka botanical name Papaver somniferum hai; immature capsule latex se opium aur morphine/codeine/papaverine jaise alkaloids milte hain. [Q58]
- Opium depressant/sedative narcotic drug category mein hai; Morphine narcotic analgesic drug hai. [Q59, Q60]

11. Tapa Cloth — Moraceae

- Tapa/bark cloth Moraceae family ke plants—source examples mulberry aur fig—ki softened inner bark se banaya jata hai. [Q61]

12. Laminaria, Iodine & Goitre

- Iodine Laminaria se obtained hota hai; Laminaria brown algae/kelp seaweed hai aur source iron/potassium richness bhi mention karta hai. [Q62]
- Iodine sea mein abundantly available substance hai aur iodine-deficiency goitre ko prevent/treat karne ke context mein administered hota hai. [Q63]